



GI70EG01V101 ECN

PGDBC0

2022.08

AUO Display+

ECN Notice



- **Purpose:**

- The original mass-produced model **G170EG01 V101** TCON IC uses NVT I2201 IVG, NVT proposed to ADP the TCON IC EC requirement, AUO Display Plus (ADP) will replace NVT I2201 IVG with Raydium TCON IC I2303 Z09 for this model. EE has confirmed that the existing structure could not find alternative materials, so **G170EG01 V104** was developed to replace it .
- And LED driver IC Richtek RT8488 is going to EOL too, ADP will replace LED driver IC RT8488 with NT50508B for this model.

- **Changes:**

- TCON IC will be changed from NVT 12201 1VG to Raydium 12303 Z09.
- Source IC will be changed from Novatek NT39633 to RM92061
- Gate IC will be changed from NT39348 to NT65208.
- LED driver IC will be changed from Richtek RT8488 to NT50508B.
- With New IC and Cell , new product specification needs to be change as below
 - a. General specification (Page3)
 - b. Backlight unit driving (Page4)
 - c. LVDS pin assignment (Page5)
 - d. LED Driver pin assignment (Page6)
 - e. Interface timing (Page7)
 - f. Backlight input signal characteristics (Page 8)
 - f. 2D Drawing –front /rear(Page 9,10)

- **Schedule**

- a. Verification sample will be available in 2022/08
- b. RA/EMI/EMC test results will be provides in 2022/10

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General Specification



Items	G170EG01 V101	G170EG01 V104—New
Screen Diagonal	17" (5:4)	17" (5:4)
Active Area [mm]	337.92(W) x 270.34(H)	337.92(W) x 270.34(H)
Resolution	1280(RGB) x 1024	1280(RGB) x 1024
Mode	TN mode, normally white	TN mode, normally white
Contrast ratio	800 : 1 (typ.)	800 : 1 (typ.)
Brightness [nits]	350 nits (typ.)	350 nits (typ.)
LED Lifetime [hours]	50K-hrs (min.)	50K-hrs (min.)
Response Time [ms] on+off	30ms (typ.)	20ms (typ.)
Viewing Angle (L/R/U/D, CR>10)	80°/80°/60°/80°	80°/80°/60°/80°
(Wx, Wy)	(0.313, 0.329)	(0.313, 0.329)
Color Gamut	NTSC 72%	NTSC 72%
Support Color	16.7 M-colors (6+ Hi FRC bit)	16.7 M-colors (8bit)
Interface LCD	LVDS	LVDS
Nominal Input Voltage [V]	5V (typ.)	5V (typ.)
LED driver board	With	With
Operation Temperature [°C]	-30 ~ 85 °C	-30 ~ 85 °C
Storage Temperature [°C]	-30 ~ 85 °C	-30 ~ 85 °C
Surface Treatment	Anti-glare	Anti-glare
Outline Dimension [mm]	358.5(W)x296.5(H)x18(D)	358.5(W)x296.5(H)x12.95(D)

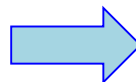
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Backlight unit driving



GI70EG01V101

Symbol	Parameter	Min.	Typ.	Max.	Unit	Remark
VLED	Input Voltage	10.8	12	12.6	Volt	
ILED	Input Current	-	1.18	-	A	100% Dimming
PLED	Power Consumption	-	14.16	-	Watt	100% Dimming
INRUSH LED	Inrush Current	-	-	5.1	A	VLED rising time ~ 470us
FPWM	PWM Dimming Frequency	200	-	20K	Hz	Note 1,2
VPWM DIM	Swing Voltage High	4.5	5.0	5.5	Volt	
	Swing Voltage Low	-	-	0.8	Volt	
DPWM	Dimming Duty Cycle	10	-	100	%	
IF	LED Forward Current	-	110	-	mA	Ta = 25°C
VLED ON/OFF	On Control Voltage	3.0	3.3	5.5	Volt	Note 3, 4
	Off Control Voltage	-	-	0.8	Volt	
Operating Life		50000	-	-	Hrs	Note 5, 6



GI70EG01 V104 -New

Symbol	Parameter	Min.	Typ.	Max.	Unit	Remark
VLED	Input Voltage	10.8	12	12.6	Volt	
ILED	Input Current	-	1.18	-	A	100% Dimming
PLED	Power Consumption	-	14.16	-	Watt	100% Dimming
INRUSH LED	Inrush Current	-	-	5.1	A	VLED rising time ~ 470us
FPWM	PWM Dimming Frequency	200	-	20K	Hz	Note 1,2
VPWM DIM	Swing Voltage High	4.5	5.0	5.5	Volt	
	Swing Voltage Low	-	-	0.8	Volt	
DPWM	Dimming Duty 200-5K Hz	5	-	100	%	
	Dimming Duty 5K-20K Hz	15	-	100	%	
IF	LED Forward Current	-	110	-	mA	Ta = 25°C
VLED ON/OFF	On Control Voltage	3.0	3.3	5.5	Volt	Note 3, 4
	Off Control Voltage	-	-	0.8	Volt	
Operating Life		50000	-	-	Hrs	Note 5, 6

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LVDS Pin Assignment

G170EG01 V101

Connector Name / Designation	Interface Connector
Manufacturer	JAE or Compatible
Type Part Number	FI-XB30SSL-HF15 or Compatible
Mating Housing Part Number	JAE FI-X30HL or Compatible

PIN #	SIGNAL NAME	PIN #	SIGNAL NAME
1	RxOIN0-	16	RxEIN1+
2	RxOIN0+	17	GND
3	RxOIN1-	18	RxEIN2-
4	RxOIN1+	19	RxEIN2+
5	RxOIN2-	20	RxECLKIN-
6	RxOIN2+	21	RxECLKIN+
7	GND	22	RxEIN3-
8	RxOCLKIN-	23	RxEIN3+
9	RxOCLKIN+	24	GND
10	RxOIN3-	25	GND (AGMODE+HVS)
11	RxOIN3+	26	GND
12	RxEIN0-	27	GND
13	RxEIN0+	28	VDD
14	GND	29	VDD
15	RxEIN1-	30	VDD

G170EG01 V104 (New)



LVDS Connector	Signal Connector
Manufacturer	STM or Compatible
Connector Model Number	MSCKT2407P30B or Compatible
Adaptable Plug	JAE FI-X30HL or Compatible

Pin no	Symbol	Pin no	Symbol
1	RxOIN0-	16	RxEIN1+
2	RxOIN0+	17	GND
3	RxOIN1-	18	RxEIN2-
4	RxOIN1+	19	RxEIN2+
5	RxOIN2-	20	RxECLKIN-
6	RxOIN2+	21	RxECLKIN+
7	GND	22	RxEIN3-
8	RxOCLKIN-	23	RxEIN3+
9	RxOCLKIN+	24	GND
10	RxOIN3-	25	GND (AGMODE+HVS)
11	RxOIN3+	26	GND
12	RxEIN0-	27	GND
13	RxEIN0+	28	VDD
14	GND	29	VDD
15	RxEIN1-	30	VDD

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LED Driver Pin Assignment



G170EG01 V101

Connector Name / Designation	Lamp Connector
Manufacturer	E&T or compatible
Connector Model Number	3806K-F06Y-03R or compatible
Mating Connector Model Number	H208K-P06N-02B or compatible

Pin#	Symbol	Signal Name
1	V_{LED}	12V
2	V_{LED}	12V
3	GND	GND
4	GND	GND
5	Display on	5V-On / 0V-Off
6	Dimming	PWM Dimming

G170EG01 V104 (New)

Connector Name / Designation	Lamp Connector
Manufacturer	STM or Compatible
Connector Model Number	MSB24038P6D or compatible
Mating Connector Model Number	H208K-P06N-02B or compatible

Pin#	Symbol	Signal Name
1	V_{LED}	12V
2	V_{LED}	12V
3	GND	GND
4	GND	GND
5	Display on	5V-On / 0V-Off
6	Dimming	PWM Dimming

Interface Timing



G170EG01 V101

Signal	Item	Symbol	Min	Typ	Max	Unit
Vertical Section	Period	Tv	1034	1066	1150	Th
	Active	Tdisp(v)	1024	1024	1024	Th
	Blanking	Tbp(v)+Tfp(v)+PWvs	10	42	126	Th
Horizontal Section	Period	Th	750	844	2048	Tclk
	Active	Tdisp(h)	640	640	640	Tclk
	Blanking	Tbp(h)+Tfp(h)+PWhs	110	204	-	Tclk
Clock	Period	Tclk	14.81	18.52	25	ns
	Frequency	Freq	40	54	70	MHz
Frame rate	Frame rate	F	49	60	76	Hz

G170EG01 V104 (New)

Signal	Item	Symbol	Min	Typ	Max	Unit
Vertical Section	Period	Tv	1032	1066	1150	Th
	Active	Tdisp(v)	1024	1024	1024	Th
	Blanking	Tbp(v)+Tfp(v)+PWvs	8	42	126	Th
Horizontal Section	Period	Th	780	844	871	Tclk
	Active	Tdisp(h)	640	640	640	Tclk
	Blanking	Tbp(h)+Tfp(h)+PWhs	140	204	231	Tclk
Clock	Period	Tclk	14.81	18.52	22.2	ns
	Frequency	Freq	45	54	67.5	MHz
Frame rate	Frame rate	F	49	60	75	Hz

Backlight input signal characteristics



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Parameter	Value			Unit
	Min.	Typ.	Max.	
T1	0.5	-	10	[ms]
T2	30	40	50	[ms]
T3	200	-	-	[ms]
T4	0.5	-	10	[ms]
T5	10	-	-	[ms]
T6	10	-	-	[ms]
T7	0	-	-	[ms]
T8	10	-	-	[ms]
T9	-	-	10	[ms]
T10	110	-	-	[ms]
T11	0	16	50	[ms]
T12	-	-	10	[ms]
T13	1000	-	-	[ms]

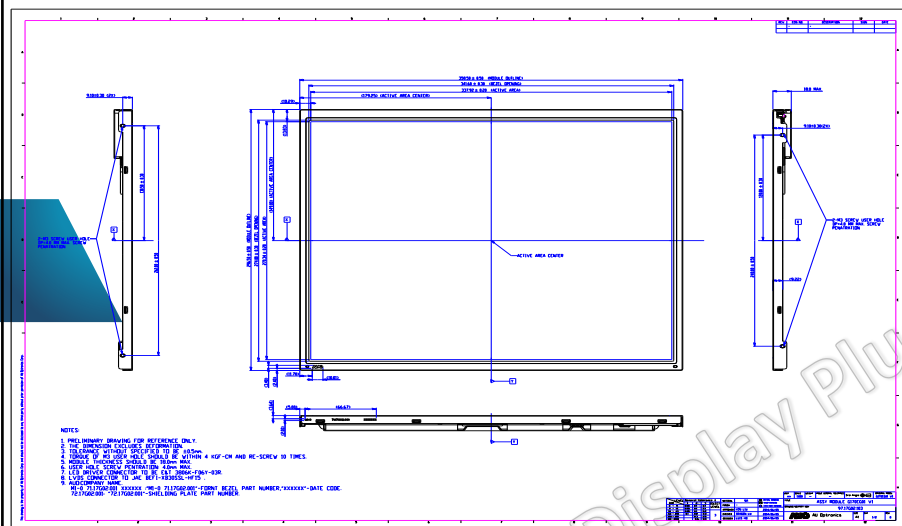
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Parameter	Value			Unit
	Min.	Typ.	Max.	
T1	0.5	--	10	[ms]
T2	30	40	50	[ms]
T3	200	--	--	[ms]
T4	0.5	--	10	[ms]
T5	10	--	--	[ms]
T6	10	--	--	[ms]
T7	0	--	--	[ms]
T8	10	--	--	[ms]
T9	--	--	10	[ms]
T10	110	--	--	[ms]
T11	0	16	50	[ms]
T12	--	--	10	[ms]
T13	1000	--	--	[ms]

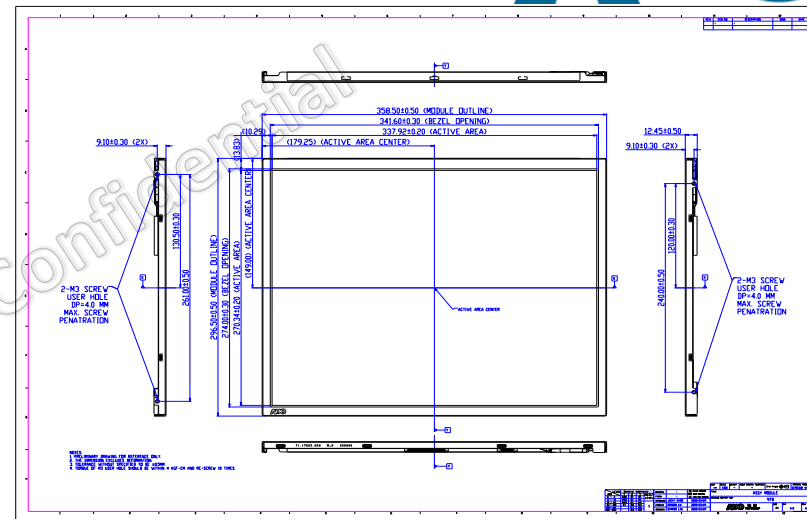
2D Drawing Comparison (Front View)



G170EG01 V101



G170EG01 V104 (New)



Comparison Table

Bezel Opening (mm)

Thickness (mm) Max

Front Bezel / Material

Mounting hole

G170EG01 V101

341.6x274

18

SECC

Side mount –Screw size ,M3
Z position 9.1mm ;
Y Position : Center to Hole 130.5 (Left side)
Center to Hole 120 (Right side)

G170EG01 V104

341.6x274

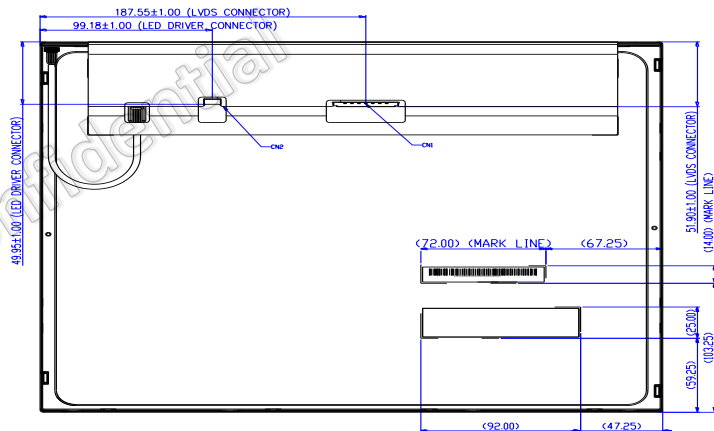
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SGLC

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